Name _____ _____ Baggett/Fall2026/207/Quiz7

first name last name

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 1 for the reason.**

1. (**2** pts) If $f'(x) < 0$ for $1 < x < 6$, then f is decreasing on $(1, 6)$.

True **False**

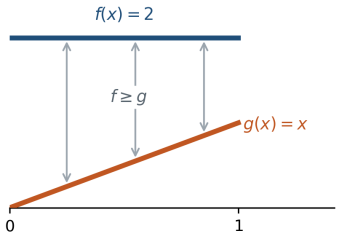
Why? The Increasing/Decreasing Test: a negative derivative throughout an interval means the function falls across the whole of it. This follows from the Mean Value Theorem, not from checking points.

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2. (**2** pts) If f and g are differentiable and $f(x) \geq g(x)$ for $a < x < b$, then $f'(x) \geq g'(x)$ for $a < x < b$.

True **False**

Why? Take $f(x) = 2$ and $g(x) = x$ on $(0, 1)$. Then $f(x) \geq g(x)$ there, but $f'(x) = 0$ and $g'(x) = 1$, so $f' < g'$ everywhere on the interval. Being larger says nothing about growing faster.



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3. (**7** pts) Let $f(x) = x^3 - 3x^2 - 9x + 5$.

(a) Find the critical numbers of f, and classify each one as a local maximum, a local minimum, or neither. **(4 points)**

$x = -1$ is a local maximum, $x = 3$ is a local minimum.

$$f'(x) = 3x^2 - 6x - 9 \qquad \text{Power Rule, term by term}$$
$$= 3\left(x^2 - 2x - 3\right) = 3(x-3)(x+1) \qquad \textbf{factor before anything else}$$
$$x = -1,\ 3 \qquad \text{a product is zero when a factor is}$$

		-1		3	
f'	$+$	0	$-$	0	$+$
test	$f'(-2) = 15$		$f'(0) = -9$		$f'(4) = 15$
f	$\nearrow$		$\searrow$		$\nearrow$

f' turns $+$ to $-$ at $x = -1$: **local max**, $f(-1) = 10$. It turns $-$ to $+$ at $x = 3$: **local min**, $f(3) = -22$.

The same answer by the Second Derivative Test. $f''(x) = 6x - 6$, and you only have to

evaluate it at the two critical numbers:

$$f''(-1) = 6(-1) - 6 = -12 < 0 \qquad \text{concave down at } x = -1 \Rightarrow \textbf{local maximum}$$
$$f''(3) = 6(3) - 6 = 12 > 0 \qquad \text{concave up at } x = 3 \Rightarrow \textbf{local minimum}$$

Either test is a complete argument — the sign chart tells you what f' does *near* each critical number, the second derivative test tells you which way the curve bends *at* it. **But** the second test is silent when $f''(c) = 0$ — which is exactly what problem 5 is about — and the first test never is.

(b) On what interval(s) is f concave up and concave down? Give the inflection point as an ordered pair. **(3 points)**

Concave down on $(-\infty, 1)$, concave up on $(1, \infty)$; inflection point $(1, -6)$.

$f''(x) = 6x - 6 = 6(x-1)$, zero only at $x = 1$.

		1	
f''	$-$	0	$+$
f	$\frown$		$\smile$

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The sign *changes* at $x = 1$, so it is an inflection point, and $f(1) = 1 - 3 - 9 + 5 = -6$ gives $(\mathbf{1}, \mathbf{-6})$.

4. (**6** pts) A function f is differentiable everywhere, and

$$f'(x) = (x+3)(x-1) \qquad \text{and} \qquad f''(x) = 2(x+1).$$

You are **not** given a formula for f.

(a) On what intervals is f increasing and decreasing, and what happens at each critical number? **(3 points)**

(b) On what intervals is f concave up and concave down, and where is the inflection point? **(3 points)**

(a) incr. on $(-\infty, -3)$, $(1, \infty)$; decr. on $(-3, 1)$; max at $x = -3$, min at $x = 1$. (b) CD on $(-\infty, -1)$, CU on $(-1, \infty)$; inflection at $x = -1$.

(a) $f' = 0$ at $x = -3,\ 1$. The parabola $(x+3)(x-1)$ opens upward, so it is positive outside the roots and negative between them.

		-3		1	
f'	$+$	0	$-$	0	$+$
f	$\nearrow$		$\searrow$		$\nearrow$

So $x = -3$ is a **local maximum** and $x = 1$ a **local minimum**.

(b) $f'' = 2(x+1)$ is zero only at $x = -1$.

		-1	
f''	$-$	0	$+$
f	$\frown$		$\smile$

The bend changes, so $x = -1$ is an **inflection point**.

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A check that costs nothing: $f''(-3) = -4 < 0$ — concave down at the local maximum, which is what a hilltop should be; $f''(1) = 4 > 0$ — concave up at the local minimum. The two charts agree.

5. (**3** pts) Suppose $f'(2) = 0$ and $f''(2) = 0$. What does the Second Derivative Test tell you about $x = 2$, and what should you do instead?

Nothing — the test is inconclusive. Use the First Derivative Test: check the sign of f' on both sides of $x = 2$.

The test only decides when $f''(c)$ is strictly positive or strictly negative; at $f''(2) = 0$ it is **silent**. All three outcomes are still possible — $y = x^4$, $y = -x^4$ and $y = x^3$ all have $f'(0) = f''(0) = 0$ at the origin, and give a minimum, a maximum and no extremum respectively.

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The fallback is the sign of f' on either side: $+$ to $-$ is a maximum, $-$ to $+$ a minimum, no change means neither.